Stat 517
Homework # 3

1. Let $Y_1 < Y_2 < Y_3 < Y_4$ be the order statistics of a random sample of size 4 from the distribution having pdf $f(x) = e^{-x}, 0 < x < \infty$, zero elsewhere. Find $P(Y_4 \geq 3)$.

   Solution: Note $Y_4 = \max\{X_1, \ldots, X_4\}$. The cdf of Y is

$$\begin{aligned} P(Y_4 \leq y) &= P(X_1 \leq y, \ldots, X_4 \leq y), \\ &= \prod_{i=1}^{4} P(X_i \leq y), \\ &= \prod_{i=1}^{4} F_{X_i}(y), \\ &= \left(1 - e^{-y}\right)^4. \end{aligned}$$

   Hence $P(Y_4 \geq 3) = 1 - P(Y_4 < 3) = 1 - P(Y_4 \leq 3) =$

$$1 - \left(1 - e^{-3}\right)^4.$$

2. Let $Y_1 < Y_2 < Y_3 < Y_4$ be the order statistics of a random sample of size $n = 4$ from the distribution

$$f(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the joint cdf of Y_3 and Y_4.

(b) Find the conditional pdf of Y_3 given $Y_4 = y_4$.

(c) Evaluate $\mathbb{E}(Y_3 \mid Y_4 = y_4)$.

Solution

The common cdf of the sample observations is

$$F(x) = \begin{cases} 0, & x \le 0, \\ x^2, & 0 < x < 1, \\ 1, & x \ge 1. \end{cases}$$

(a) The joint cdf is

$$G_{Y_3,Y_4}(y_3, y_4) = P(Y_3 \le y_3,\ Y_4 \le y_4).$$

First, suppose that $0 < y_3 < y_4 < 1$. The event $\{Y_4 \le y_4\}$ means that all four observations are at most y_4. In addition, $\{Y_3 \le y_3\}$ means that at least three of the four observations are at most y_3.

Therefore, either all four observations are at most y_3, or exactly three observations are at most y_3 and the remaining observation lies in $(y_3, y_4]$. Hence,

$$\begin{aligned} G_{Y_3,Y_4}(y_3, y_4) &= [F(y_3)]^4 + \binom{4}{3}[F(y_3)]^3\big[F(y_4) - F(y_3)\big] \\ &= (y_3^2)^4 + 4(y_3^2)^3(y_4^2 - y_3^2) \\ &= 4y_3^6y_4^2 - 3y_3^8. \end{aligned}$$

If $0 < y_4 < 1$ and $y_3 \ge y_4$, then the condition $Y_3 \le y_3$ is automatically satisfied whenever $Y_4 \le y_4$. Thus,

$$G_{Y_3,Y_4}(y_3, y_4) = P(Y_4 \le y_4) = [F(y_4)]^4 = y_4^8.$$

If $0 < y_3 < 1$ and $y_4 \ge 1$, then the condition $Y_4 \le y_4$ is automatic, and therefore

$$\begin{aligned} G_{Y_3,Y_4}(y_3, y_4) &= P(Y_3 \le y_3) \\ &= \binom{4}{3}[F(y_3)]^3[1 - F(y_3)] + [F(y_3)]^4 \\ &= 4y_3^6(1 - y_3^2) + y_3^8 \\ &= 4y_3^6 - 3y_3^8. \end{aligned}$$

Consequently, the joint cdf is

$$G_{Y_3,Y_4}(y_3,y_4) = \begin{cases} 0, & y_3 \le 0 \text{ or } y_4 \le 0, \\ 4y_3^6y_4^2 - 3y_3^8, & 0 < y_3 < y_4 < 1, \\ y_4^8, & 0 < y_4 < 1,\ y_3 \ge y_4, \\ 4y_3^6 - 3y_3^8, & 0 < y_3 < 1,\ y_4 \ge 1, \\ 1, & y_3 \ge 1,\ y_4 \ge 1. \end{cases}$$

For $0 < y_3 < y_4 < 1$, the corresponding joint pdf is

$$\begin{aligned} g_{Y_3,Y_4}(y_3,y_4) &= \frac{\partial^2}{\partial y_3\,\partial y_4} G_{Y_3,Y_4}(y_3,y_4) \\ &= \frac{\partial^2}{\partial y_3\,\partial y_4}\left(4y_3^6y_4^2 - 3y_3^8\right) \\ &= 48y_3^5y_4. \end{aligned}$$

Thus,

$$g_{Y_3,Y_4}(y_3,y_4) = \begin{cases} 48y_3^5y_4, & 0 < y_3 < y_4 < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(b) The marginal pdf of Y_4 is

$$\begin{aligned} g_{Y_4}(y_4) &= \int_0^{y_4} g_{Y_3,Y_4}(y_3,y_4)\,dy_3 \\ &= \int_0^{y_4} 48y_3^5y_4\,dy_3 \\ &= 48y_4\left[\frac{y_3^6}{6}\right]_0^{y_4} \\ &= 8y_4^7, \qquad 0 < y_4 < 1. \end{aligned}$$

Therefore, the conditional pdf of Y_3 given $Y_4 = y_4$ is

$$\begin{aligned} g_{Y_3|Y_4}(y_3 \mid y_4) &= \frac{g_{Y_3,Y_4}(y_3,y_4)}{g_{Y_4}(y_4)} \\ &= \frac{48y_3^5y_4}{8y_4^7} \\ &= \frac{6y_3^5}{y_4^6}. \end{aligned}$$

Hence,

$$g_{Y_3|Y_4}(y_3 \mid y_4) = \begin{cases} \dfrac{6y_3^5}{y_4^6}, & 0 < y_3 < y_4, \\ 0, & \text{otherwise,} \end{cases} \qquad 0 < y_4 < 1.$$

(c) The conditional expectation is

$$\begin{aligned}
\mathbb{E}(Y_3 \mid Y_4 = y_4) &= \int_0^{y_4} y_3 \, g_{Y_3|Y_4}(y_3 \mid y_4)\, dy_3 \\
&= \int_0^{y_4} y_3 \frac{6y_3^5}{y_4^6}\, dy_3 \\
&= \frac{6}{y_4^6} \int_0^{y_4} y_3^6\, dy_3 \\
&= \frac{6}{y_4^6} \left[\frac{y_3^7}{7}\right]_0^{y_4} \\
&= \frac{6}{7} y_4.
\end{aligned}$$

Therefore,

$$\boxed{\mathbb{E}(Y_3 \mid Y_4 = y_4) = \frac{6}{7} y_4}.$$

3. Let X_1, X_2, X_3 be a random sample from a continuous-type distribution having pdf $f(x) = 2x$, $0 < x < 1$, zero elsewhere.

   (a) Compute the probability that the smallest of X_1, X_2, X_3 exceeds the median of the distribution.

   (b) If $Y_1 < Y_2 < Y_3$ are the order statistics, find the correlation between Y_2 and Y_3.

Solution:

(a) The CDF is $F(x) = x^2$ for $0 < x < 1$. The median m can be computed by

$$\int_0^m 2x dx = \frac{1}{2} \Rightarrow x^2 = \frac{1}{2} \Rightarrow x = \frac{\sqrt{2}}{2}.$$

Thus,

$$P(X_{(1)} > m) = P(X_{(1)} > \frac{\sqrt{2}}{2}) = [1 - (\frac{\sqrt{2}}{2})^2]^3 = \frac{1}{8}.$$

(b) By formula (5.2.3), the joint PDF of Y_2 and Y_3 is

$$\begin{aligned} g_{23}(y_2, y_3) =& \frac{3!}{(2-1)!(3-2-1)!(3-3)!}(y_2^2)^{2-1}(y_3 - y_2)^{3-2-1}(1-y_3)^{3-3}(2y_2)(2y_3) \\ =& 24y_2^3 y_3 \end{aligned}$$

for $0 < y_2 < y_3 < 1$. Then, we have

$$\begin{aligned} E(Y_2) =& \int_0^1 [\int_0^{y_3} y_2 g_{23}(y_2, y_3) dy_2] dy_3 \\ =& 24 \int_0^1 [\int_0^{y_3} y_2^4 dy_2] y_3 dy_3 \\ =& \frac{24}{5} \int_0^1 y_3^6 dy_3 \\ =& \frac{24}{35}; \end{aligned}$$

$$\begin{aligned} E(Y_2^2) =& \int_0^1 [\int_0^{y_3} y_2^2 g_{23}(y_2, y_3) dy_2] dy_3 \\ =& 24 \int_0^1 [\int_0^{y_3} y_2^5 dy_2] y_3 dy_3 \\ =& 4 \int_0^1 y_3^7 dy_3 \\ =& \frac{1}{2}. \end{aligned}$$

Then,

$$V(Y_2) = \frac{1}{2} - (\frac{24}{35})^2 = 0.0298.$$

In addition,

$$
\begin{aligned}
E(Y_3) =& \int_0^1 [\int_0^{y_3} y_3 g_{23}(y_2, y_3) dy_2] dy_3 \\
=& 24 \int_0^1 [\int_0^{y_3} y_2^3 dy_2] y_3^2 dy_3 \\
=& 6 \int_0^1 y_3^6 dy_3 \\
=& \frac{6}{7};
\end{aligned}
$$

$$
\begin{aligned}
E(Y_3^2) =& \int_0^1 [\int_0^{y_3} y_3^2 g_{23}(y_2, y_3) dy_2] dy_3 \\
=& 24 \int_0^1 [\int_0^{y_3} y_2^3 dy_2] y_3^3 dy_3 \\
=& 6 \int_0^1 y_3^7 dy_3 \\
=& \frac{3}{4};
\end{aligned}
$$

Then,

$$V(Y_3) = \frac{3}{4} - (\frac{6}{7})^2 = 0.0153.$$

Also, we have

$$
\begin{aligned}
E(Y_2 Y_3) =& \int_0^1 [\int_0^{y_3} y_2 y_3 g_{23}(y_2, y_3) dy_2] dy_3 \\
=& 24 \int_0^1 [\int_0^{y_3} y_2^4 dy_2] y_3^2 dy_3 \\
=& \frac{24}{5} \int_0^1 y_3^7 dy_3 \\
=& \frac{3}{5}.
\end{aligned}
$$

Therefore,

$$Cov(Y_2, Y_3) = \frac{3}{5} - \frac{24}{35} \times \frac{6}{7} = 0.0122$$

and

$$Corr(Y_2, Y_3) = \frac{0.0122}{\sqrt{0.0298 \times 0.0153}} = 0.573.$$

4. Let $Y_1 < Y_2 < Y_3 < Y_4 < Y_5$ denote the order statistics of a random sample of size 5 from a distribution having pdf $f(x) = e^{-x}$, $0 < x < \infty$, zero elsewhere. Show that $Z_1 = Y_2$ and $Z_2 = Y_4 - Y_2$ are independent.

   Solution: Here $F(x) = 1 - e^{-x}$, so $1 - F(y) = e^{-y}$ and $F(y_4) - F(y_2) = e^{-y_2} - e^{-y_4}$. The joint pdf of Y_2 and Y_4 $(j = 2,\ k = 4,\ n = 5)$ is

$$\begin{aligned} g_{2,4}(y_2, y_4) &= \frac{5!}{(2-1)!\,(4-2-1)!\,(5-4)!}\,[F(y_2)]^1\,[F(y_4) - F(y_2)]^1\,[1 - F(y_4)]^1\,f(y_2)f(y_4) \\ &= 120\,(1 - e^{-y_2})(e^{-y_2} - e^{-y_4})\,e^{-y_2}\,e^{-2y_4}, \qquad 0 < y_2 < y_4 < \infty. \end{aligned}$$

   Let $U = Y_2$ and $V = Y_4 - Y_2$, so that $y_2 = u$, $y_4 = u + v$, with Jacobian 1. Substituting,

$$\begin{aligned} (1 - e^{-y_2}) &\to (1 - e^{-u}), \\ (e^{-y_2} - e^{-y_4}) &\to e^{-u} - e^{-(u+v)} = e^{-u}(1 - e^{-v}), \\ e^{-y_2} &\to e^{-u}, \\ e^{-2y_4} &\to e^{-2(u+v)} = e^{-2u}e^{-2v}. \end{aligned}$$

   Therefore the joint pdf of U and V is

$$\begin{aligned} g_{U,V}(u, v) &= 120\,(1 - e^{-u})\,e^{-u}(1 - e^{-v})\,e^{-u}\,e^{-2u}e^{-2v} \\ &= 120\,(1 - e^{-u})\,e^{-4u}\,(1 - e^{-v})\,e^{-2v} \\ &= \underbrace{\left[20(1 - e^{-u})e^{-4u}\right]}_{h(u)}\ \underbrace{\left[6(1 - e^{-v})e^{-2v}\right]}_{k(v)}, \qquad 0 < u < \infty,\ 0 < v < \infty. \end{aligned}$$

   Since the joint pdf factors as a function of u alone times a function of v alone over a rectangular support, $U = Y_2$ and $V = Y_4 - Y_2$ are independent.

   (Check: $\int_0^\infty 20(1 - e^{-u})e^{-4u}\,du = 20\left(\frac{1}{4} - \frac{1}{5}\right) = 1$ and $\int_0^\infty 6(1 - e^{-v})e^{-2v}\,dv = 6\left(\frac{1}{2} - \frac{1}{3}\right) = 1$, so h and k are the marginal densities of U and V.)

5. A die was cast $n = 120$ independent times and the following data resulted: If we use a chi-

Sports Up	1	2	3	4	5	6
Frequency	b	20	20	20	20	40 - b

square test, for what values of b would the hypothesis that the die is unbiased be rejected at the 0.025 significance level?

Solution: The hypothesis is

$$H_0 : \text{unbiased versus } H_1 : \text{ not } H_0.$$

The chi-square statistic at the significance level of 0.025 with degree 5 is approximately 12.83. Therefore, the following statistic has to greater than the number to reject the null hypothesis. That is

$$Q = \frac{(b-20)^2}{20} + \frac{(20-20)^2}{20} \times 4 + \frac{(40-b-20)^2}{20} > 12.83.$$

Hence, when $b \geq 32$ or $b \leq 8$, we can conclude that the die is not unbiased.

6. Two different teaching procedures were used on two different groups of students. Each group contained 100 students of about the same ability. At the end of the term, an evaluation team assigned a letter grade to each student:

	Grade					
Group	A	B	C	D	F	Total
I	15	25	32	17	11	100
II	9	18	29	28	16	100
Total	24	43	61	45	27	200

Test at the 5% significance level the hypothesis that the two grade distributions are the same (i.e. the two teaching procedures are equally effective).

Solution: Let X_{ij} be the observed count in row i, column j. Under the hypothesis of homogeneity, the expected counts are

$$\hat{X}_{ij} = \frac{(\text{row total}_i)(\text{column total}_j)}{n}.$$

Since both row totals equal 100 and $n = 200$, every expected count is simply $\hat{X}_{ij} = (\text{column total}_j)/2$, the same for both rows:

	Grade				
Group	A	B	C	D	F
I	12	21.5	30.5	22.5	13.5
II	12	21.5	30.5	22.5	13.5

The Pearson chi-square statistic is

$$\chi^2 = \sum_{i=1}^{2}\sum_{j=1}^{5} \frac{(X_{ij} - \hat{X}_{ij})^2}{\hat{X}_{ij}} = 2\left[\frac{(15-12)^2}{12} + \frac{(25-21.5)^2}{21.5} + \frac{(32-30.5)^2}{30.5} + \frac{(17-22.5)^2}{22.5} + \frac{(11-13.5)^2}{13.5}\right]$$
$$= 2\,(0.750 + 0.570 + 0.074 + 1.344 + 0.463)$$
$$= 2\,(3.201) \approx 6.40,$$

where the two rows contribute equally because their observed−expected differences are equal in magnitude.

The degrees of freedom are

$$(r-1)(c-1) = (2-1)(5-1) = 4,$$

and the critical value is $\chi^2_{0.05,\,4} = 9.488$. Since

$$\chi^2 \approx 6.40 < 9.488,$$

we fail to reject H_0. There is not enough evidence that the two grade distributions differ, so the two teaching procedures appear to be equally effective.

7. Prove the converse of the Theorem of Monte Carlo. That is, let X be a random variable with a continuous cdf $F(x)$. Assume that $F(x)$ is strictly increasing on the space of X. Consider the random variable $Z = F(X)$. Show that Z has a uniform distribution on the interval $(0, 1)$.

   Solution: Suppose that there exists an inverse function of F. The cdf of Z is

$$\begin{aligned} P(Z \leq z) &= P(F(X) \leq z) \\ &= P\left(X \leq F^{-1}(z)\right) \\ &= F\left(F^{-1}(z)\right) \\ &= z, \end{aligned}$$

   which is the cdf of a uniform distribution.

8. Suppose we are interested in a particular Weibull distribution with pdf

$$f(x) = \begin{cases} \dfrac{1}{\theta^3} 3x^2 e^{-x^3/\theta^3} & 0 < x < \infty \\ 0 & \text{otherwise.} \end{cases}$$

Determine a method to generate random observations from this Weibull distribution.

Solution: The cdf of X is

$$\begin{aligned} F(x) &= \int_0^x \frac{1}{\theta^3} 3t^2 e^{-t^3/\theta^3} dt \quad (\text{Let } y = t^3/\theta^3) \\ &= \int_0^{x^3/\theta^3} e^{-y} dy \\ &= 1 - e^{-x^3/\theta^3}. \end{aligned}$$

Since this cdf is strictly increasing, we can find $F^{-1}(u)$,

$$F^{-1}(u) = x = \theta \left(\ln \left(\frac{1}{1-u} \right) \right)^{1/3}.$$

These are steps to get samples of X:

(1) Generate $U \sim \text{Unif}(0, 1)$.

(2) Evaluate $X = F^{-1}(U)$